

EKUAZIO-SISTEMAK

2018 UZTAILA B1

a) Eztabaidatu $S(a)$ sistema a parametroaren arabera:

$$S(a) = \begin{cases} x + ay - z = 2 \\ 2x + y + az = 0 \\ 3x + (a+1)y - z = a-1 \end{cases}$$

b) Ba al dago soluzioarik $a = 1$ baliorako? Erantzuna baiezkoa bada, eman soluzioa. Ezezkoa bada, erantzuna arrazoitu.

2018 EKAINA B1

$S(a)$ sistema hau emanda:

$$S(a) = \begin{cases} x + 2y - z = 2 \\ x + (a+1)y - az = 2a \\ x + ay + (a+1)z = 1 \end{cases}$$

a) Eztabaidatu sistema a parametroaren arabera.

b) Soluziorik al dago $a = 2$ baliorako? Erantzuna baiezkoa bada, kalkulatu soluzioa. Ezezkoa bada, arrazoitu erantzuna.

2017 UZTAILA A1

Eztabaidatu sistema hau a parametroaren balioen arabera:

$$ax + 2y + 6z = 0$$

$$2x + ay + 4z = 2$$

$$2x + ay + 6z = a - 2$$

Existitzen baldin bada, aurkitu soluzioa $a = 0$ kasurako.

2017 EKAINA A1

Eztabaidatu sistema hau m parametroaren balioen arabera (EZ da beharrezkoa ebaztea)

$$2x + y - z = 1$$

$$x + my + z = 2$$

$$3x + y - mz = 3$$

2016 UZTAILA A1

Eztabaidatu nola jokutzen duen sistema honek b parametroaren balioen arabera.

$$x + y + z = 0$$

$$-x + 2y + bz = -3$$

$$x - 2y - z = b$$

Aurkitu soluzioa, baldin eta soluzioa badu, $b = 2$ kasurako.

2016 EKAINA A1

Eztabaidatu sistema hau b parametroaren balioen arabera (EZ da inola ere beharrezkoa ebaztea).

$$x + 2y - z = 2$$

$$x + (1 + b)y - bz = 2b$$

$$x + by + (1 + b)z = 1$$

2015 UZTAILA A1

Eztabaidatu ekuazio-sistema m parametroaren arabera

$$mx + my = 1$$

$$3x + mz = m - 2$$

$$-y + z = m - 3$$

Ba al da indeterminazio-kasurik? Erantzuna baiezkoa bada, ebatzi sistema kasu horietan. Ezezkoa bada, azaldu zergatia.

2015 EKAINA B1

Ekuazio linealen sistema hau emanda:

$$x + y - z = -4$$

$$3x + ay + z = a - 1$$

$$2x + ay = -2$$

- Eztabaidatu nola jokutzen duen sistemak a parametroaren balioen arabera.
- Ebatzi sistema indeterminazio-kasuan edo -kasuetan.
- Ba al da a -ren baliorik sistemak soluziorik ez izatea dakarrenik? Arrazoitu erantzuna.

2014 UZTAILA A1

Ekuazio linealen sistema hau emanda:

$$-x + my + 2z = m$$

$$2x + my - z = 2$$

$$mx - y + 2z = m$$

- a) Deskriba ezazu nola jokutzen duen sistemak m parametroaren balioen arabera.
- b) $m = -1$ kasuan, ebatz ezazu sistema, ahal baldin bada. Ezinezkoa bada, azaldu ezazu zergatia.

2014 EKAINA A1

Ekuazio linealen sistema hau emanda:

$$x - 2y - z = -1$$

$$ax - y + 2z = 2$$

$$x + 2y + az = 3$$

- a) Eztabaidatu ezazu nola jokutzen duen sistemak a parametroaren balioen arabera.
- b) Ebatzi ezazu sistema soluzio bat baino gehiago dituenean.

2013 UZTAILA A1

Ekuazio-sistema hau emanda:

$$\begin{cases} x + y + az = 1 \\ ay - z = -1 \\ x + 2ay = 0 \end{cases}$$

- a) Egin eztabaida a -ren balioen arabera.
- b) Ebatzi sistema, bateragarri indeterminatua denean.

2013 EKAINA B1

Sistema hau emanda:

$$\begin{cases} mx + my + 2z = m \\ x + (m-2)y = -1 \\ 2y + 2z = 2 \end{cases}$$

- a) Egin eztabaida m parametroaren balioen arabera.
- b) Ebatz ezazu, ahal bada, $m = 5$ kasurako.

2012 ÜZTAILA A1

Sistema hau emanda:

$$\begin{cases} (m-1)x + y + z = m \\ x + (m-1)y + z = 0 \\ y + z = 1 \end{cases}$$

- a) Eztabaidatu ezazu m parametroaren balioaren arabera.
- b) Ebatz ezazu, ahal bada, $m = 0$ eta $m = 3$ kasuetarako.

2012 EKAINA A1

Sistema hau emanda:

$$\begin{cases} x + (A+1)y + Az = A+1 \\ Ay + z = 0 \\ x + y = 1 \end{cases}$$

- a) Eztabaidatu ezazu A parametroaren balioaren arabera.
- b) Ebatz ezazu, ahal bada, $A = 4$ kasurako.

SELEKTIBILITATEA: ECUAZII-SISTEMAK

2018 UZTALA B1)

a)
$$S = \begin{cases} x + ay - z = 2 \\ 2x + y + az = 0 \\ 3x + (a+1)y - z = a-1 \end{cases} \quad n=3 \text{ (cubunul Karna)}$$

$$A = \begin{pmatrix} 1 & a & -1 \\ 2 & 1 & a \\ 3 & a+1 & -1 \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & a & -1 & 2 \\ 2 & 1 & a & 0 \\ 3 & a+1 & -1 & a-1 \end{pmatrix}$$

$$|A| = 0 \Rightarrow -1 - 2(a+1) + 3a^2 + 3 - a(a+1) + 2a = 0$$

$$-1 - 2a - 2 + 3a^2 + 3 - a^2 - a + 2a = 0 \Rightarrow 2a^2 - a = 0 \Rightarrow a(2a-1) = 0$$

$$\begin{cases} a=0 \\ a=\frac{1}{2} \end{cases}$$

• $a \neq 0$ et $a \neq \frac{1}{2}$ denean $\Rightarrow h(A) = 3 = h(A') = n \Rightarrow$ Sistema bazeagun determinatua

• $a=0$ denean: $h(A) < 3$

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 3 & 1 & -1 \end{pmatrix}$$

$$\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} \neq 0 \Rightarrow h(A) = 2$$

$$(A') \rightarrow \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 1 & -1 \end{vmatrix}$$

$$\Rightarrow -1 + 4 - 6 \neq 0 \Rightarrow h(A') = 3$$

$h(A) \neq h(A')$
Sistema bazeazina

• $a = \frac{1}{2}$ denean

$$A = \begin{pmatrix} 1 & 1/2 & -1 \\ 2 & 1 & 1/2 \\ 3 & 3/2 & -1 \end{pmatrix}$$

$$\begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix} \neq 0 \Rightarrow h(A) = 2$$

$$A' = \begin{pmatrix} 1 & 1/2 & -1 & 2 \\ 2 & 1 & 1/2 & 0 \\ 3 & 3/2 & -1 & -1/2 \end{pmatrix}$$

$$\begin{vmatrix} 1 & -1 & 2 \\ 2 & 1/2 & 0 \\ 3 & -1 & -1/2 \end{vmatrix} = -\frac{1}{4} - 4 - 3 - 1 \neq 0 \Rightarrow h(A') = 3$$

$a = \frac{1}{2}$ denean sistema bazeazina da

b) $a=1$ denean $\rightarrow |A| \neq 0 \rightarrow h(A) = 3 = h(A') = n$

Sistema batzagarri determinatua

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & -1 \end{pmatrix} \quad |A| = 1$$

Cramer:

$$x = \frac{\begin{vmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & -1 \end{vmatrix}}{1} = \frac{-6}{1} \rightarrow \boxed{x = -6}$$

$$A' = \begin{pmatrix} 1 & 1 & -1 & 2 \\ 2 & 1 & 1 & 0 \\ 3 & 2 & -1 & 0 \end{pmatrix}$$

$$y = \frac{\begin{vmatrix} 1 & 2 & -1 \\ 2 & 0 & 1 \\ 3 & 0 & -1 \end{vmatrix}}{1} = \frac{6+4}{1} \rightarrow \boxed{y = 10}$$

$$\text{Soluzioa: } \begin{cases} x = -6 \\ y = 10 \\ z = 2 \end{cases}$$

$$z = \frac{\begin{vmatrix} 1 & 1 & 2 \\ 2 & 1 & 0 \\ 3 & 2 & 0 \end{vmatrix}}{1} = \frac{8-6}{1} \rightarrow \boxed{z = 2}$$

(soluzioetan txarto)

2018 EKATIMA B1

$$S = \begin{cases} x + 2y - z = 2 \\ x + (a+1)y - az = 2a \\ x + ay + (a+1)z = 1 \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & a+1 & -a \\ 1 & a & a+1 \end{pmatrix}$$

a)

$$A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & a+1 & -a & 2a \\ 1 & a & a+1 & 1 \end{pmatrix}$$

$$|A| = 0 \rightarrow (a+1)^2 - a - 2a + a + 1 + a^2 - 2(a+1) = 0$$

$$a^2 + 2a + 1 - 2a + 1 + a^2 - 2a - 2 = 0 \rightarrow 2a^2 - 2a = 0$$

$$a^2 - a = 0 \rightarrow a(a-1) = 0 \rightarrow \begin{cases} a = 0 \\ a = 1 \end{cases}$$

• $a \neq 0$ eta $a \neq 1$ denean $\rightarrow h(A) = 3 = h(A') \Rightarrow$ Sistema batzagarri determinatua

• $a=0$ dengan

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

(2)

$$\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \neq 0 \rightarrow h(A) = 2$$

$$A' \Rightarrow \begin{vmatrix} 1 & 2 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 1 - 2 - 2 \neq 0 \rightarrow h(A') = 3$$

$\left. \begin{array}{l} h(A) \neq h(A') \\ \text{Sistema batereane} \end{array} \right\}$

• $a=1$ dengan

$$A = \begin{pmatrix} \textcircled{1} & \textcircled{2} & -1 \\ 1 & 2 & -1 \\ \textcircled{1} & \textcircled{1} & 2 \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & 2 & -1 & 2 \\ 1 & 1 & 2 & 1 \end{pmatrix}$$

$$\textcircled{A} \rightarrow \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \neq 0 \rightarrow h(A) = 2$$

$$\textcircled{A'} \rightarrow \begin{vmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 1 & 1 & 1 \end{vmatrix} = 0 \rightarrow h(A') = 2$$

$\left. \begin{array}{l} 2 = h(A) = h(A') < n = 3 \\ \text{Sistema bateregan indeterminate} \end{array} \right\}$

b) $a=2 \rightarrow$ Sistem bateregan determinatua.

$|A| = 4$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & -2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & 3 & -2 & 4 \\ 1 & 2 & 3 & 1 \end{pmatrix}$$

Cramer-en erregela

$$x = \frac{\begin{vmatrix} 2 & 2 & -1 \\ 4 & 3 & -2 \\ 1 & 2 & 3 \end{vmatrix}}{|A|} = \frac{6 - 13}{4} \rightarrow \boxed{x = -7/4}$$

$$y = \frac{\begin{vmatrix} 1 & 2 & -1 \\ 1 & 4 & -2 \\ 1 & 1 & 3 \end{vmatrix}}{|A|} = \frac{7 - 0}{4} \Rightarrow \boxed{y = 7/4}$$

$$z = \frac{\begin{vmatrix} 1 & 2 & 2 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix}}{|A|} = \frac{15 - 16}{4} \Rightarrow$$

$$\rightarrow \boxed{z = -1/4}$$

Soluzioa $\begin{cases} x = -7/4 \\ y = 7/4 \\ z = -1/4 \end{cases}$

$$\begin{cases} ax + 2y + 6z = 0 \\ 2x + ay + 4z = 2 \\ 2x + ay + 6z = a-2 \end{cases}$$

$n=3$ (etragun Kop)

$$A = \begin{pmatrix} a & 2 & 6 \\ 2 & a & 4 \\ 2 & a & 6 \end{pmatrix} \quad A' = \begin{pmatrix} a & 2 & 6 & 0 \\ 2 & a & 4 & 2 \\ 2 & a & 6 & a-2 \end{pmatrix}$$

$$|A|=0 \rightarrow 6a^2 + 12a + 16 - 12a - 4a^2 - 24 = 0 \Rightarrow 2a^2 - 8 = 0$$

$$a^2 = 4 \Rightarrow \begin{cases} a=2 \\ a=-2 \end{cases}$$

• $a \neq 2$ eta $a \neq -2$ denean $\rightarrow h(A)=3=h(A')=n$

Sistema bateragarri determinatua

• $a=2$ denean:

$$A = \begin{pmatrix} 2 & 2 & 6 \\ 2 & 2 & 4 \\ 2 & 2 & 6 \end{pmatrix} \quad A' = \begin{pmatrix} 2 & 2 & 6 & 0 \\ 2 & 2 & 4 & 2 \\ 2 & 2 & 6 & 0 \end{pmatrix}$$

(A) $\rightarrow \begin{vmatrix} 2 & 6 \\ 2 & 4 \end{vmatrix} \neq 0 \rightarrow h(A)=2$

(A') $\rightarrow \begin{vmatrix} 2 & 6 & 0 \\ 2 & 4 & 2 \\ 2 & 6 & 0 \end{vmatrix} = 24 - 24 = 0 \rightarrow h(A')=2$

$h(A)=h(A')=2 < 3=n$

Sistema bateragarri indeterminatua

• $a=-2$ denean:

$$A = \begin{pmatrix} -2 & 2 & 6 \\ 2 & -2 & 4 \\ 2 & -2 & 6 \end{pmatrix} \quad A' = \begin{pmatrix} -2 & 2 & 6 & 0 \\ 2 & -2 & 4 & 2 \\ 2 & -2 & 6 & -4 \end{pmatrix}$$

(A) $\rightarrow \begin{vmatrix} 2 & 6 \\ -2 & 4 \end{vmatrix} \neq 0 \rightarrow h(A)=2$

(A') $\rightarrow \begin{vmatrix} 2 & 6 & 0 \\ -2 & 4 & 2 \\ -2 & 6 & -4 \end{vmatrix} = -32 - 24 - 24 - 48 \neq 0 \rightarrow h(A')=3$

$h(A) \neq h(A')$

Sistema bateraezina

$a=0$ dan $a=0$ → Sistemanya determinat

$$A = \begin{pmatrix} 0 & 2 & 6 \\ 2 & 0 & 4 \\ 2 & 0 & 6 \end{pmatrix} \quad A' = \begin{pmatrix} 0 & 2 & 6 & 0 \\ 2 & 0 & 4 & 2 \\ 2 & 0 & 6 & -2 \end{pmatrix}$$

$|A| = -8$

Cramer-en erregela aplikatu

$$x = \frac{\begin{vmatrix} 0 & 2 & 6 \\ 2 & 0 & 4 \\ -2 & 0 & 6 \end{vmatrix}}{|A|} = \frac{-16 - 24}{-8} = \frac{-40}{-8} \rightarrow \boxed{x = 5}$$

$$y = \frac{\begin{vmatrix} 0 & 0 & 6 \\ 2 & 2 & 4 \\ 2 & -2 & 6 \end{vmatrix}}{|A|} = \frac{-24 - 24}{-8} = \frac{-48}{-8} \rightarrow \boxed{y = 6}$$

$$z = \frac{\begin{vmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 2 & 0 & -2 \end{vmatrix}}{|A|} = \frac{8 - (-8)}{-8} = \frac{16}{-8} \rightarrow \boxed{z = -2}$$

Soluzioa $\begin{cases} x = 5 \\ y = 6 \\ z = -2 \end{cases}$

2017 EKAINA A1

$$\begin{cases} 2x + y - z = 1 \\ x + my + z = 2 \\ 3x + y - mz = 3 \end{cases}$$

$h(A) \leq 3$

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & m & 1 \\ 3 & 1 & -m \end{pmatrix}$$

$h(A') \leq 3$

$$A' = \begin{pmatrix} 2 & 1 & -1 & 1 \\ 1 & m & 1 & 2 \\ 3 & 1 & -m & 3 \end{pmatrix}$$

① → $\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2$

② $n=3$ (erretagun Kopurua)

$$\begin{vmatrix} 2 & 1 & -1 \\ 1 & m & 1 \\ 3 & 1 & -m \end{vmatrix} = 0 \rightarrow -2m^2 - 1 + 3 + 3m - 2 + m = 0 \rightarrow -2m^2 + 4m = 0$$

$$2m(-m + 2) = 0 \rightarrow \begin{cases} m = 0 \\ m = 2 \end{cases}$$

• $m \neq 0$ eta $m \neq 2$ denean $\rightarrow h(A)=3=h(A')=n \Rightarrow$ Sistema bateragari determinatua

• $m=0$ denean

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 0 & 1 \\ 3 & 1 & 0 \end{pmatrix}$$

$$A' = \begin{pmatrix} 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & 2 \\ 3 & 1 & 0 & 3 \end{pmatrix}$$

① $\rightarrow h(A)=2$

② $\rightarrow \begin{vmatrix} 2 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 3 \end{vmatrix} = 7 - 7 = 0 \rightarrow h(A')=2$

$\left. \begin{array}{l} h(A) = h(A') = 2 < 3 = n \\ \text{Sistema bateragari indeterminatua} \end{array} \right\}$

• $m=2$ denean

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{pmatrix}$$

$$A' = \begin{pmatrix} 2 & 1 & -1 & 1 \\ 1 & 2 & 1 & 2 \\ 3 & 1 & -2 & 3 \end{pmatrix}$$

① $\rightarrow h(A)=2$

② $\rightarrow \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{vmatrix} = 19 - 13 \neq 0 \rightarrow h(A')=3$

$\left. \begin{array}{l} h(A) \neq h(A') \\ \text{Sistema bateragari ezina} \end{array} \right\}$

2016 UZTAI LA A1

$$\begin{cases} x + y + z = 0 \\ -x + 2y + bz = -3 \\ x - 2y - z = b \end{cases}$$

$$n=3$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 2 & b \\ 1 & -2 & -1 \end{pmatrix}$$

$$h(A) \leq 3$$

$$A' = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & b & -3 \\ 1 & -2 & -1 & b \end{pmatrix}$$

$$h(A') \leq 3$$

$$\textcircled{A} \rightarrow \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2$$

$$\begin{vmatrix} 1 & 1 & 1 \\ -1 & 2 & b \\ 1 & -2 & -1 \end{vmatrix} = 0 \rightarrow -2 + 2 + b - 2 + 2b - 1 = 0 \rightarrow 3b - 3 = 0 \rightarrow b = 1$$

$$\bullet \text{ } \boxed{b \neq 1 \text{ denean}} \quad h(A) = 3 = h(A') = n \rightarrow \boxed{\text{Sistema bateragarri determinatua}}$$

$$\bullet \text{ } \boxed{b = 1 \text{ denean}} \quad A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 2 & 1 \\ 1 & -2 & -1 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & 1 & -3 \\ 1 & -2 & -1 & 1 \end{pmatrix}$$

$$\textcircled{A} \rightarrow \boxed{h(A) = 2}$$

$$\textcircled{A'} \rightarrow \begin{vmatrix} 1 & 1 & 0 \\ -1 & 2 & -3 \\ 1 & -2 & 1 \end{vmatrix} = -1 - 5 \neq 0 \rightarrow \boxed{h(A') = 3}$$

$$\left. \begin{array}{l} h(A) \neq h(A') \\ \text{Sistema bateraezina} \end{array} \right\}$$

b=2 dengan $\rightarrow$ Sistema batasan determinat ($h(A) = h(A') = 3 = n$)

$$\begin{cases} x + y + z = 0 \\ -x + 2y + 2z = -3 \\ x - 2y - z = 2 \end{cases}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 2 & 2 \\ 1 & -2 & -1 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & 2 & -3 \\ 1 & -2 & -1 & 2 \end{pmatrix}$$

$$\hookrightarrow |A| = 3$$

Cramer-en erregela

$$x = \frac{\begin{vmatrix} 0 & 1 & 1 \\ -3 & 2 & 2 \\ 2 & -2 & -1 \end{vmatrix}}{|A|} = \frac{10 - 7}{3} \Rightarrow \boxed{x = 1}$$

$$y = \frac{\begin{vmatrix} 1 & 0 & 1 \\ -1 & -3 & 2 \\ 1 & 2 & -1 \end{vmatrix}}{|A|} = \frac{1 - 1}{3} = \frac{0}{3} \rightarrow \boxed{y = 0}$$

$$z = \frac{\begin{vmatrix} 1 & 1 & 0 \\ -1 & 2 & -3 \\ 1 & -2 & 2 \end{vmatrix}}{|A|} = \frac{1 - 4}{3} = \frac{-3}{3} \rightarrow \boxed{z = -1}$$

Soluzioa:

$$\begin{cases} x = 1 \\ y = 0 \\ z = -1 \end{cases}$$

2016 EKAINA A1

$$\begin{cases} x + 2y - z = 2 \\ x + (1+b)y - bz = 2b \\ x + by + (1+b)z = 1 \end{cases}$$

$n=3$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1+b & -b \\ 1 & b & 1+b \end{pmatrix}$$

$h(A) \leq 3$

$$A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & 1+b & -b & 2b \\ 1 & b & 1+b & 1 \end{pmatrix}$$

$h(A') \leq 3$

(A) $\rightarrow |A|=0 \rightarrow \begin{vmatrix} 1 & 2 & -1 \\ 1 & 1+b & -b \\ 1 & b & 1+b \end{vmatrix} = 0 \rightarrow (1+b)^2 - b - 2b + (1+b) + b^2 - 2(1+b) = 0$

$\rightarrow 1 + 2b + b^2 - 3b + 1 + b + b^2 - 2 - 2b = 0$
 $2b^2 - 2b = 0 \rightarrow 2b(b-1) = 0 \rightarrow \begin{cases} b=0 \\ b=1 \end{cases}$

• $b \neq 0$ eta $b \neq 1$ denean $\rightarrow h(A) = h(A') = 3 = n \rightarrow$ Sistema bateragarri determinatua

• $b=0$ denean $\rightarrow A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}$

(A) $\rightarrow \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \neq 0 \rightarrow h(A)=2$ (ezin da 3 izan)

(A') $\rightarrow \begin{vmatrix} 1 & 2 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 1 - 4 \neq 0 \rightarrow h(A')=3$

$\left. \begin{matrix} h(A) \neq h(A') \\ h(A)=2, h(A')=3 \end{matrix} \right\} \rightarrow$ Sistema batezina

• $b=1$ denean $\rightarrow A = \begin{pmatrix} \textcircled{1} & \textcircled{2} & -1 \\ 1 & 2 & -1 \\ \textcircled{1} & \textcircled{1} & 2 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 1 & 2 & -1 & 2 \\ 1 & 1 & 2 & 1 \end{pmatrix}$

(A) $\rightarrow \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \neq 0 \rightarrow h(A)=2$

(A') $\rightarrow \begin{vmatrix} \textcircled{1} & \textcircled{2} & 2 \\ 1 & 2 & 2 \\ \textcircled{1} & \textcircled{1} & 1 \end{vmatrix} = 0 \rightarrow h(A')=2$

$\left. \begin{matrix} h(A)=h(A')=2 < 3=n \end{matrix} \right\} \rightarrow$ Sistema bateragarri indeterminatua



$$\begin{cases} mx + my = 1 \\ 3x + mz = m-2 \\ -y + z = m-3 \end{cases}$$

$$n=3$$

$$A = \begin{pmatrix} m & m & 0 \\ 3 & 0 & m \\ 0 & -1 & 1 \end{pmatrix}$$

$$h(A) \leq 3$$

$$A' = \begin{pmatrix} m & m & 0 & 1 \\ 3 & 0 & m & m-2 \\ 0 & -1 & 1 & m-3 \end{pmatrix}$$

$$h(A') \leq 3$$

$$\textcircled{A} \rightarrow \begin{vmatrix} 3 & 0 \\ 0 & -1 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2 \text{ beti}$$

$$|A|=0 \rightarrow \begin{vmatrix} m & m & 0 \\ 3 & 0 & m \\ 0 & -1 & 1 \end{vmatrix} = 0 \rightarrow 0 + m^2 - 3m = 0 \rightarrow m^2 - 3m = 0$$

$$m(m-3)=0 \rightarrow \begin{cases} m=0 \\ m=3 \end{cases}$$

$$\bullet \boxed{m \neq 0 \text{ eta } m \neq 3} \text{ denean } \rightarrow h(A)=3 = h(A')=n \rightarrow \boxed{\text{Sistema bateragarri determinatua}}$$

$$\bullet \boxed{m=0} \text{ denean } \rightarrow A = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 0 & -1 & 1 \end{pmatrix} \quad A' = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & -2 \\ 0 & -1 & 1 & -3 \end{pmatrix}$$

$$\textcircled{A} \rightarrow \boxed{h(A)=2}$$

$$\textcircled{A'} \rightarrow \begin{vmatrix} 0 & 0 & 1 \\ 3 & 0 & -2 \\ 0 & -1 & -3 \end{vmatrix} = -3 \neq 0 \rightarrow \boxed{h(A')=3}$$

$$\left. \begin{array}{l} h(A) \neq h(A') \\ \end{array} \right\} \boxed{\text{Sistema bateraezina}}$$

$$\bullet \boxed{m=3} \text{ denean } \rightarrow A = \begin{pmatrix} 3 & 3 & 0 \\ 3 & 0 & 3 \\ 0 & -1 & 1 \end{pmatrix} \quad A' = \begin{pmatrix} 3 & 3 & 0 & 1 \\ 3 & 0 & 3 & 1 \\ 0 & -1 & 1 & 0 \end{pmatrix}$$

$$\textcircled{A} \rightarrow \boxed{h(A)=2}$$

$$\textcircled{A'} \rightarrow \begin{vmatrix} 3 & 3 & 1 \\ 3 & 0 & 1 \\ 0 & -1 & 0 \end{vmatrix} = -3 - (-3) = 0 \rightarrow \boxed{h(A')=2}$$

$$\left. \begin{array}{l} h(A) = 2 = h(A') < 3 = n \\ \end{array} \right\} \boxed{\text{Sistema bateragarri indeterminatua}}$$

Bai → indeterminasi Kosua $m=3$ denean: $h(A) = h(A') = 2$

$$\begin{cases} 3x + 3y = 1 \\ 3x + 3z = 1 \\ -y + z = 0 \end{cases}$$

Bi ekuasio dagot linealki independentak, beraz bat kendu daiteke.

$$\begin{cases} 3x + 3z = 1 \rightarrow 3x = 1 - 3z \\ -y + z = 0 \rightarrow y = z \end{cases} \rightarrow \begin{cases} x = \frac{1}{3} - z \\ y = z \end{cases}$$

$$z = \lambda \text{ eginet}$$

$$\text{Soluzioa: } \begin{cases} x = \frac{1}{3} - \lambda \\ y = \lambda \\ z = \lambda \end{cases} \quad \lambda \in \mathbb{R}$$

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$$\begin{cases} x + y - z = -4 \\ 3x + ay + z = a-1 \\ 2x + ay = -2 \end{cases}$$

$n=3$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 3 & a & 1 \\ 2 & a & 0 \end{pmatrix}$$

$$h(A) \leq 3$$

$$A' = \begin{pmatrix} 1 & 1 & -1 & -4 \\ 3 & a & 1 & a-1 \\ 2 & a & 0 & -2 \end{pmatrix}$$

$$h(A') \leq 3$$

a) $(A) \rightarrow \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2$ beti

$$|A| = 0 \rightarrow \begin{vmatrix} 1 & 1 & -1 \\ 3 & a & 1 \\ 2 & a & 0 \end{vmatrix} = 0 \rightarrow -3a + 2 + 2a - a = 0 \rightarrow 2 - 2a = 0$$

$$2a = 2 \rightarrow a = 1$$

• $a \neq 1$ denean $\rightarrow h(A) = 3 = h(A') = n \rightarrow$ Sistema bateragarri determinatua

• $a = 1$ denean

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 3 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & -1 & -4 \\ 3 & 1 & 1 & 0 \\ 2 & 1 & 0 & -2 \end{pmatrix}$$

$(A) \rightarrow h(A) = 2$

$(A') \rightarrow \begin{vmatrix} 1 & -1 & -4 \\ 3 & 1 & 0 \\ 2 & 0 & -4 \end{vmatrix} = -4 - (-4) = 0 \rightarrow h(A') = 2$

$a = 1$ denean $\rightarrow$
 $\rightarrow h(A) = h(A') = 2 < 3 = n$
 Sistema bateragarri indeterminatua

b) $a=1$ deanean $\rightarrow h(A) = h(A') = 2 \rightarrow$ 2 ekuazio linealki independente dagoz $\rightarrow$ Bat Kendu duteke

$$\begin{cases} x + y - z = -4 \\ 3x + y + z = 0 \\ 2x + y = -2 \end{cases}$$

$$\boxed{z = \lambda} \text{ egokor}$$

$$\underline{\underline{\lambda \in \mathbb{R}}}$$

$$2x + y = -2 \rightarrow \boxed{y = -2 - 2x} \rightarrow y = -2 - 2(2 - \lambda)$$

$$\boxed{y = 2\lambda - 6}$$

$$x - 2 - 2x - \lambda = -4 \rightarrow -x = \lambda - 2 \rightarrow \boxed{x = 2 - \lambda}$$

$$\boxed{\text{Soluzioa: } \begin{cases} x = 2 - \lambda \\ y = 2\lambda - 6 \\ z = \lambda \end{cases}}$$

$$\underline{\underline{\lambda \in \mathbb{R}}}$$

c) a) atalean ikusi dugunaren arabera kasu guztietan sistema bateragarria da $h(A) = h(A')$ delako a parametraren edozein balioetarako. Beraz ez dago kasurik sistemak soluziorik ez daukanik.

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$$\begin{cases} -x + my + 2z = m \\ 2x + my - z = 2 \\ mx - y + 2z = m \end{cases}$$

$$\boxed{n=3}$$

$$A = \begin{pmatrix} -1 & m & 2 \\ 2 & m & -1 \\ m & -1 & 2 \end{pmatrix}$$

$$A' = \begin{pmatrix} -1 & m & 2 & m \\ 2 & m & -1 & 2 \\ m & -1 & 2 & m \end{pmatrix}$$

a) $\textcircled{A} \rightarrow \begin{vmatrix} -1 & 2 \\ 2 & -1 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2$ beti.

$$|A| = 0 \rightarrow \begin{vmatrix} -1 & m & 2 \\ 2 & m & -1 \\ m & -1 & 2 \end{vmatrix} = 0 \rightarrow -2m - 4 - m^2 - 2m^2 + 1 - 4m = 0$$

$$-3m^2 - 6m - 3 = 0 \rightarrow m^2 + 2m + 1 = 0 \rightarrow (m+1)^2 = 0 \rightarrow \boxed{m = -1}$$

• $m \neq -1$ denetan $\rightarrow h(A) = 3 = h(A') = n \rightarrow$ Sistema bateragarri determinatua

• $m = -1$ denetan

$$A = \begin{pmatrix} -1 & -1 & 2 \\ 2 & -1 & -1 \\ -1 & -1 & 2 \end{pmatrix} \quad A' = \begin{pmatrix} -1 & -1 & 2 & -1 \\ 2 & -1 & -1 & 2 \\ -1 & -1 & 2 & -1 \end{pmatrix}$$

(A) $\rightarrow h(A) = 2$

(A') $\rightarrow \begin{vmatrix} -1 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & -1 \end{vmatrix} = -9 - (-9) = 0 \rightarrow h(A') = 2$

gainera
2 errenkada berdinak dira

$h(A) = h(A') = 2 < 3 = n$
Sistema bateragarri indeterminatua

b) $m = -1$

$$\begin{cases} -x - y + 2z = -1 \\ 2x - y - z = 2 \\ -x - y + 2z = -1 \end{cases}$$

$(z = \lambda)$ eginet $\lambda \in \mathbb{R}$

$2(1+z) - y - z = 2 \rightarrow 2 + 2z - z - 2 = y$

$-x - y = -1 - 2z$

$2x - y = 2 + z$

$-3x = -3 - 3z \rightarrow x = 1 + z$

$y = z$

Beraz,

Soluzioa $\begin{cases} x = 1 + \lambda \\ y = \lambda \\ z = \lambda \end{cases} \quad \lambda \in \mathbb{R}$

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$$\begin{cases} x - 2y - z = -1 \\ ax - y + 2z = 2 \\ x + 2y + az = 3 \end{cases}$$

$$A = \begin{pmatrix} \textcircled{1} & \textcircled{-2} & -1 \\ a & -1 & 2 \\ \textcircled{1} & \textcircled{2} & a \end{pmatrix}$$

$$A' = \begin{pmatrix} 1 & -2 & -1 & -1 \\ a & -1 & 2 & 2 \\ 1 & 2 & a & 3 \end{pmatrix}$$

$n=3$

a) $\textcircled{A} \rightarrow \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix} = 2+2=4 \neq 0 \rightarrow h(A) \geq 2$ beti

$$|A|=0 \rightarrow \begin{vmatrix} 1 & -2 & -1 \\ a & -1 & 2 \\ 1 & 2 & a \end{vmatrix} = 0 \rightarrow -a - 2a - 4 - 1 - 4 + 2a^2 = 0$$

$$2a^2 - 3a - 9 = 0 \begin{cases} a=3 \\ a=-\frac{3}{2} \end{cases}$$

$\bullet$ $a \neq 3$ eta $a \neq -\frac{3}{2}$ denean $\rightarrow h(A) = h(A') = 3 = n \rightarrow$ Sistema bateragarri determinatua

$\bullet$ $a=3$ denean $\rightarrow A = \begin{pmatrix} \textcircled{1} & \textcircled{-2} & -1 \\ 3 & -1 & 2 \\ \textcircled{1} & \textcircled{2} & 3 \end{pmatrix} \quad A' = \begin{pmatrix} \textcircled{1} & \textcircled{-2} & -1 & -1 \\ 3 & -1 & 2 & 2 \\ \textcircled{1} & \textcircled{2} & 3 & 3 \end{pmatrix}$

$\textcircled{A} \rightarrow h(A)=2$

$\textcircled{A'} \rightarrow \begin{vmatrix} 1 & -2 & -1 \\ 3 & -1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 0 \rightarrow h(A')=2$

$\left. \begin{array}{l} h(A) = h(A') = 2 < 3 = n \\ \text{Sistema bateragarri indeterminatua} \end{array} \right\}$

• $a = \frac{-3}{2}$ denean $\rightarrow$ $A = \begin{pmatrix} \textcircled{1} & \textcircled{-2} & -1 \\ -3/2 & -1 & 2 \\ \textcircled{1} & \textcircled{2} & -3/2 \end{pmatrix}$ $A' = \begin{pmatrix} \textcircled{1} & \textcircled{-2} & -1 & -1 \\ -3/2 & -1 & 2 & 2 \\ \textcircled{1} & \textcircled{2} & -3/2 & 3 \end{pmatrix}$ (8)

$\textcircled{A} \rightarrow \boxed{h(A) = 2}$

$\textcircled{A'} \rightarrow \begin{vmatrix} 1 & -2 & -1 \\ -3/2 & -1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -4 - 14 \neq 0 \rightarrow \boxed{h(A') = 3}$

$h(A) \neq h(A')$

Sistema
baterazena

b) Sistema baterazari indeterminatua da $\rightarrow$ $\boxed{a=3}$ denean $\boxed{z=\lambda}$ eginet $\lambda \in \mathbb{R}$

$$\begin{cases} x - 2y - z = -1 \\ 3x - y + 2z = 2 \\ \cancel{x + 2y + 3z = 3} \end{cases} \quad \begin{cases} x - 2y = z - 1 \xrightarrow{\cdot 3} 3x - 6y = 3z - 3 \\ 3x - y = -2z + 2 \xrightarrow{-} \underline{3x - y = -2z + 2} \\ -5y = 5z - 5 \end{cases}$$

Lehenengo ekuazioa

$x - 2(-z + 1) - z = -1$

$x + 2z - 2 - z = -1$

$\boxed{x = 1 - z}$

$\boxed{y = -z + 1}$

$\boxed{y = 1 - z}$

Soluzioa :

$$\begin{cases} x = 1 - \lambda \\ y = 1 - \lambda \\ z = \lambda \end{cases} \quad \underline{\lambda \in \mathbb{R}}$$

$$\begin{cases} x + y + az = 1 \\ ay - z = -1 \\ x + 2ay = 0 \end{cases}$$

$n=3$

$h(A) \leq 3$

$$A = \begin{pmatrix} 1 & 1 & a \\ 0 & a & -1 \\ 1 & 2a & 0 \end{pmatrix}$$

$h(A') \leq 3$

$$A' = \begin{pmatrix} 1 & 1 & a & 1 \\ 0 & a & -1 & -1 \\ 1 & 2a & 0 & 0 \end{pmatrix}$$

a) $(A) \rightarrow |A|=0 \rightarrow \begin{vmatrix} 1 & 1 & a \\ 0 & a & -1 \\ 1 & 2a & 0 \end{vmatrix} = 0 \rightarrow -1 - a^2 + 2a = 0$

$-a^2 + 2a - 1 = 0 \rightarrow a^2 - 2a + 1 = 0 \rightarrow (a-1)^2 = 0 \rightarrow \boxed{a=1}$

• $\boxed{a \neq 1}$ denean $\Rightarrow h(A) = h(A') = 3 = n \rightarrow$ Sistema bateragari determinatua

• $\boxed{a=1}$ denean $\rightarrow A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 0 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & -1 \\ 1 & 2 & 0 & 0 \end{pmatrix}$

$(A) \rightarrow \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} \neq 0 \rightarrow \boxed{h(A)=2}$

$(A') \rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 0 \end{vmatrix} = -1 - (-1) = 0 \rightarrow \boxed{h(A')=2}$

$\left. \begin{array}{l} h(A) = h(A') = 2 < 3 = n \\ \end{array} \right\}$ Sistema bateragari indeterminatua

b) $\boxed{a=1}$ denean $\begin{cases} x + y + z = 1 \\ y - z = -1 \\ x + 2y = 0 \end{cases} \rightarrow \begin{cases} y = z - 1 \\ x = 2 - 2z \end{cases}$ 2 ekuazio linealki independente

$\boxed{z = \lambda}$

$x + (z-1) + z = 1 \rightarrow \boxed{x = 2 - 2z}$

Soluzioa: $\begin{cases} x = 2 - 2\lambda \\ y = \lambda - 1 \\ z = \lambda \end{cases}, \lambda \in \mathbb{R}$

$$\begin{cases} mx + my + 2z = m \\ x + (m-2)y = -1 \\ 2y + 2z = 2 \end{cases}$$

$$n=3$$

$$A = \begin{pmatrix} m & m & 2 \\ 1 & m-2 & 0 \\ 0 & 2 & 2 \end{pmatrix} \quad A' = \begin{pmatrix} m & m & 2 & m \\ 1 & m-2 & 0 & -1 \\ 0 & 2 & 2 & 2 \end{pmatrix}$$

a) $(A) \rightarrow \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2$ beti

$$|A| = 0 \rightarrow \begin{vmatrix} m & m & 2 \\ 1 & m-2 & 0 \\ 0 & 2 & 2 \end{vmatrix} = 0 \rightarrow 2m(m-2) + 4 - 2m = 0$$

$$2m^2 - 4m + 4 - 2m = 0 \rightarrow 2m^2 - 6m + 4 = 0 \rightarrow m^2 - 3m + 2 = 0$$

$$\begin{matrix} \swarrow \\ m=1 \\ \searrow \\ m=2 \end{matrix}$$

• $m \neq 2$ eta $m \neq 1$ denean $\rightarrow h(A) = h(A') = 3 = n \rightarrow$ Sistema bateragarri determinatua

• $m=2$ denean $A = \begin{pmatrix} 2 & 2 & 2 \\ 1 & 0 & 0 \\ 0 & 2 & 2 \end{pmatrix} \quad A' = \begin{pmatrix} 2 & 2 & 2 & 2 \\ 1 & 0 & 0 & -1 \\ 0 & 2 & 2 & 2 \end{pmatrix}$

$(A) \rightarrow h(A) = 2$

$(A') \rightarrow \begin{vmatrix} 2 & 2 & 2 \\ 1 & 0 & -1 \\ 0 & 2 & 2 \end{vmatrix} = 4 - 0 \neq 0 \rightarrow h(A') = 3$

$\left. \begin{matrix} h(A) \neq h(A') \\ \text{Sistema bateragarria} \end{matrix} \right\}$

• $m=1$ denean $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 0 & 2 & 2 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & -1 & 0 & -1 \\ 0 & 2 & 2 & 2 \end{pmatrix}$

$(A) \rightarrow h(A) = 2$

$(A') \rightarrow \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 0 & 2 & 2 \end{vmatrix} = 2 - 2 = 0 \rightarrow h(A') = 2$

$\left. \begin{matrix} h(A) = h(A') = 2 < 3 = n \\ \text{Sistema bateragarri indeterminatua} \end{matrix} \right\}$

b) $m=5$

$$\begin{cases} 5x + 5y + 2z = 5 \\ x + 3y = -1 \\ 2y + 2z = 2 \end{cases}$$

$$A = \begin{pmatrix} 5 & 5 & 2 \\ 1 & 3 & 0 \\ 0 & 2 & 2 \end{pmatrix} \quad A' = \begin{pmatrix} 5 & 5 & 2 & 5 \\ 1 & 3 & 0 & -1 \\ 0 & 2 & 2 & 2 \end{pmatrix}$$

$\hookrightarrow |A| = 24$

Cramer-en erregela

$$x = \frac{\begin{vmatrix} 5 & 5 & 2 \\ -1 & 3 & 0 \\ 2 & 2 & 2 \end{vmatrix}}{|A|} = \frac{26 - 2}{24} \rightarrow \boxed{x = 1}$$

$$y = \frac{\begin{vmatrix} 5 & 5 & 2 \\ 1 & -1 & 0 \\ 0 & 2 & 2 \end{vmatrix}}{|A|} = \frac{-6 - 10}{24} = \frac{-16}{24} \rightarrow \boxed{y = -\frac{2}{3}}$$

$$z = \frac{\begin{vmatrix} 5 & 5 & 5 \\ 1 & 3 & -1 \\ 0 & 2 & 2 \end{vmatrix}}{|A|} = \frac{40 - 0}{24} = \frac{40}{24} \rightarrow \boxed{z = \frac{5}{3}}$$

Soluzioa :

$$\begin{cases} x = 1 \\ y = -\frac{2}{3} \\ z = \frac{5}{3} \end{cases}$$

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$$\begin{cases} (m-1)x + y + z = m \\ x + (m-1)y + z = 0 \\ y + z = 1 \end{cases}$$

$n=3$

$$A = \begin{pmatrix} m-1 & 1 & 1 \\ 1 & m-1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$A' = \begin{pmatrix} m-1 & 1 & 1 & m \\ 1 & m-1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

a) $(A) \rightarrow \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} \neq 0 \rightarrow h(A) \geq 2$ beti

$$|A|=0 \rightarrow \begin{vmatrix} m-1 & 1 & 1 \\ 1 & m-1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 0 \rightarrow (m-1)^2 + 1 - (m-1) - 1 = 0 \rightarrow$$

$$\rightarrow m^2 - 2m + 1 - m + 1 = 0 \rightarrow m^2 - 3m + 2 = 0 \rightarrow \begin{cases} m=1 \\ m=2 \end{cases}$$

• $m \neq 1$ eta $m \neq 2$ denean $\rightarrow h(A) = h(A') = 3 = n \rightarrow$ Sistema bateragarri determinatua

• $m=1$ denean $\rightarrow A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad A' = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$

$(A) \rightarrow h(A) = 2$

$(A') \rightarrow \begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 1 - 1 = 0 \rightarrow h(A') = 2$

$h(A) = 2 = h(A') < 3 = n$
Sistema bateragarri indeterminatua

• $m=2$ denean $\rightarrow A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$

$(A) \rightarrow \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} \neq 0 \rightarrow h(A) = 2$

$(A') \rightarrow \begin{vmatrix} 1 & 1 & 2 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 3 - 1 = 2 \neq 0 \rightarrow h(A') = 3$

$h(A) \neq h(A')$
Sistema bateraezina

b) $m=0$ Kasuan ebatzi daiteke

$$\begin{cases} -x + y + z = 0 \\ x - y + z = 0 \\ y + z = 1 \end{cases}$$

$$(A|A') = \left(\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right) \rightarrow |A| = 2$$

Cramer-en erregela aplikatu

$$x = \frac{\begin{vmatrix} 0 & 1 & 1 \\ 0 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix}}{|A|} = \frac{1 - (-1)}{2} \rightarrow \boxed{x=1}$$

$$y = \frac{\begin{vmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix}}{|A|} = \frac{1 - (-1)}{2} \rightarrow \boxed{y=1}$$

$$z = \frac{\begin{vmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{vmatrix}}{|A|} = \frac{1 - 1}{2} \rightarrow \boxed{z=0}$$

$$\text{Soluzioa} \begin{cases} x=1 \\ y=1 \\ z=0 \end{cases}$$

$m=3$ Kasuan ere ebatzi daiteke

$$\begin{cases} 2x + y + z = 3 \\ x + 2y + z = 0 \\ y + z = 1 \end{cases}$$

$$(A|A') = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right) \rightarrow |A| = 2$$

Cramer-en erregela aplikatu

$$x = \frac{\begin{vmatrix} 3 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix}}{|A|} = \frac{7 - 5}{2} \rightarrow \boxed{x=1}$$

$$y = \frac{\begin{vmatrix} 2 & 3 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix}}{|A|} = \frac{1 - 5}{2} \rightarrow \boxed{y=-2}$$

$$z = \frac{\begin{vmatrix} 2 & 1 & 3 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{vmatrix}}{|A|} = \frac{7 - 1}{2} \rightarrow \boxed{z=3}$$

$$\text{Soluzioa} \begin{cases} x=1 \\ y=-2 \\ z=3 \end{cases}$$

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$$\begin{cases} x + (a+1)y + az = a+1 \\ \quad \quad \quad ay + z = 0 \\ x + y \quad \quad = 1 \end{cases}$$

$$A = \begin{pmatrix} 1 & a+1 & a \\ 0 & a & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & a+1 & a & a+1 \\ 0 & a & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

$$n=3$$

$$a) |A|=0 \rightarrow \begin{vmatrix} 1 & a+1 & a \\ 0 & a & 1 \\ 1 & 1 & 0 \end{vmatrix} = 0 \rightarrow a+1 - a^2 - 1 = 0 \rightarrow -a^2 + a = 0 \rightarrow$$

$$\rightarrow a(-a+1)=0 \begin{cases} a=0 \\ a=1 \end{cases}$$

$$\bullet \boxed{a \neq 0 \text{ eta } a \neq 1} \text{ denean} \rightarrow h(A) = h(A') = 3 = n \rightarrow$$

Sistema
bateragarri
determinatua

$$\bullet \boxed{a=0} \text{ denean} \rightarrow A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

$$\textcircled{A} \rightarrow \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0 \rightarrow \boxed{h(A)=2}$$

$$\textcircled{A'} \rightarrow \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 0 \rightarrow \boxed{h(A')=2}$$

$$h(A) = h(A') = 2 < 3 = n$$

Sistema bateragarri
indeterminatua

$$\bullet \boxed{a=1} \text{ denean} \rightarrow A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad A' = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

$$\textcircled{A} \rightarrow \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \neq 0 \rightarrow \boxed{h(A)=2}$$

$$\textcircled{A'} \rightarrow \begin{vmatrix} 1 & 2 & 2 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 1 - 2 = -1 \neq 0 \rightarrow \boxed{h(A')=3}$$

$$h(A) \neq h(A')$$

Sistema
bateraezina

b) $a=4$ denean $\rightarrow$ Ebatzi duteke

$$\begin{cases} x + 5y + 4z = 5 \\ 4y + z = 0 \\ x + y = 1 \end{cases}$$

$$(A|A') = \left(\begin{array}{ccc|c} 1 & 5 & 4 & 5 \\ 0 & 4 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right)$$

$$\rightarrow |A| = 5 - 17 = \underline{\underline{-12}}$$

Cramer-en erregela erabiliz

$$x = \frac{\begin{vmatrix} 5 & 5 & 4 \\ 0 & 4 & 1 \\ 1 & 1 & 0 \end{vmatrix}}{|A|} = \frac{5 - 21}{-12} = \frac{-16}{-12} \rightarrow \boxed{x = \frac{4}{3}}$$

$$y = \frac{\begin{vmatrix} 1 & 5 & 4 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}}{|A|} = \frac{5 - 1}{-12} = \frac{4}{-12} \rightarrow \boxed{y = -\frac{1}{3}}$$

$$z = \frac{\begin{vmatrix} 1 & 5 & 5 \\ 0 & 4 & 0 \\ 1 & 1 & 1 \end{vmatrix}}{|A|} = \frac{4 - 20}{-12} = \frac{-16}{-12} \rightarrow \boxed{z = \frac{4}{3}}$$

$$\boxed{\text{Soluzioa} \begin{cases} x = \frac{4}{3} \\ y = -\frac{1}{3} \\ z = \frac{4}{3} \end{cases}}$$